Gateway Coding-Model Eval

verifiedScore × complexity band (LOW / MEDIUM / HIGH) · ExecuteSpec V2 BUILD pipeline, PROTOTYPE mode

2026-06-18 | branch feat/mistral-coding-models | ENTERPRISE per-run model pin | 40/42 cells valid

deepseek-v4-flash
VALUE CHAMPION · 100/94/94 · $0.004–0.11

kimi-k2.7-code
BEST OPEN MODEL · 100/97/94

HIGH band
THE REAL DIFFERENTIATOR · 94 → 39

$32
TOTAL LLM SPEND, 40 RUNS

verifiedScore matrix

MODEL	TYPE	LOW	MED	HIGH	READ
claude-sonnet	base	100	97	91	Baseline (Anthropic). 100/97/91.
gpt-5.4	base	100	94	80	Baseline (OpenAI). 100/94/80.
gemini-flash	base	100	100	79	Baseline (Google). 100/100/79 — dips at HIGH.
kimi-k2.7-code	gateway	100	97	94	★ Strongest open model — 100/97/94, holds up at HIGH.
deepseek-v4-flash	gateway	100	94	94	★ Value champion — 100/94/94 at $0.004–0.11/run (~25× cheaper than frontier).
minimax-m3	gateway	100	67	91	100/67/91 — erratic (weak MED), weak caching (11%), slow.
grok-4.20	gateway	100	100	87	100/100/87 — strong, but token-hungry (12.9M in / $3+).
devstral-2	gateway	75	85	75	75/85/75 — skips tests → capped at 75 (PROMPT-FIXABLE).
codestral-2508	gateway	97	67	67	97/67/67 — capability ceiling at complexity.
glm-5.2	gateway	100	100	46	100/100/46 — collapses at HIGH.
minimax-m2.7	gateway	100	94	41	100/94/41 — collapses at HIGH.
deepseek-v4-pro	gateway	100	100	39	100/100 at low/med, COLLAPSES at HIGH (39, 3/6 tasks).
gpt-5.3-codex	gateway	100	100	–	100/100 at low/med; HIGH incomplete (OpenRouter credits).
qwen3-coder-next	gateway	75	43	–	75/43; HIGH invalid (OpenRouter credits). Weak + skips tests.

Colour: ■ ≥90 ■ 70–89 ■ <70 ■ N/A. Two HIGH cells (gpt-5.3-codex, qwen3-coder-next) are N/A — OpenRouter credits exhausted mid-run.

Key findings

- **deepseek-v4-flash is the value standout of the entire field.** 100/94/94 across bands at **$0.004 / $0.016 / $0.11** per run — roughly 25× cheaper than the frontier baselines ($2.8–3.1 at HIGH). Its HIGH score (94) ties the best open model and beats every native baseline at HIGH (gpt 80, gemini 79).
- **HIGH complexity is where models separate.** At LOW almost everything scores 100; at HIGH the field spreads from 94 (kimi, ds-flash) down to ~40 (glm-5.2, minimax-m2.7, deepseek-v4-pro).
- **Three models collapse at HIGH:** deepseek-v4-pro (100→39, 3/6 tasks), glm-5.2 (100→46), minimax-m2.7 (94→41). Strong at low/medium, but they fail to carry multi-task HIGH specs to completion.
- **Two PROMPT-FIXABLE gaps (not capability):** devstral-2 and qwen3-coder-next produce clean, building, reviewer-CLEAN, accuracy-100 code but **skip tests** → testCoverage≈10 → the no-coverage trust cap pins them at 75. Both are coding-specialised open models that under-follow the shared "tests-mandatory" directive. A per-model prompt variant should recover ~25 pts.
- **grok-4.20 is strong (100/100/87) but token-hungry** — 12.9M input tokens / $3+ — a real efficiency outlier vs glm-5.2/gpt-5.3-codex which stay lean.

Full detail — score · tasks · tokens · cost · latency

MODEL	BAND	SCORE	TASKS	IN TOK	OUT TOK	COST $	LAT (S)	PIN
claude-sonnet	low	100	2/2	15k	7k	0.325	139	n/a
claude-sonnet	medium	97	4/4	38k	19k	0.828	437	n/a
claude-sonnet	high	91	6/6	253k	63k	2.847	2673	n/a
gpt-5.4	low	100	3/3	142k	2k	0.262	65	n/a
gpt-5.4	medium	94	4/4	407k	14k	0.844	398	n/a
gpt-5.4	high	80	4/4	1.75M	42k	3.081	573	n/a
gemini-flash	low	100	2/2	309k	8k	0.172	140	n/a
gemini-flash	medium	100	4/4	835k	31k	0.802	404	n/a
gemini-flash	high	79	4/4	2.59M	63k	2.998	814	n/a
kimi-k2.7-code	low	100	2/2	149k	4k	0.299	105	ok
kimi-k2.7-code	medium	97	1/1	178k	6k	0.436	205	ok
kimi-k2.7-code	high	94	7/7	1.47M	41k	2.146	1319	ok
deepseek-v4-flash	low	100	1/1	127k	3k	0.004	69	ok
deepseek-v4-flash	medium	94	1/1	529k	19k	0.016	290	ok
deepseek-v4-flash	high	94	5/5	3.35M	114k	0.109	2134	ok
minimax-m3	low	100	2/2	152k	6k	0.222	308	ok
minimax-m3	medium	67	0/1	64k	4k	0.147	1468	ok
minimax-m3	high	91	0/1	1.36M	24k	1.331	1827	ok
grok-4.20	low	100	3/3	334k	4k	0.368	158	ok
grok-4.20	medium	100	3/4	7.87M	37k	2.846	853	ok
grok-4.20	high	87	7/7	5.37M	58k	3.061	2067	ok
devstral-2	low	75	1/1	69k	2k	0.121	59	ok
devstral-2	medium	85	4/4	396k	8k	0.584	639	ok
devstral-2	high	75	7/7	1.43M	40k	1.534	1251	ok
codestral-2508	low	97	0/1	326k	4k	0.163	113	ok
codestral-2508	medium	67	0/1	64k	4k	0.129	154	ok
codestral-2508	high	67	1/1	123k	5k	0.147	252	ok
glm-5.2	low	100	1/1	78k	2k	0.176	92	ok
glm-5.2	medium	100	1/1	190k	11k	0.443	497	ok
glm-5.2	high	46	5/6	1.13M	35k	2.109	2897	ok
minimax-m2.7	low	100	2/2	265k	6k	0.266	203	ok
minimax-m2.7	medium	94	1/1	588k	24k	0.400	609	ok
minimax-m2.7	high	41	3/4	1.10M	31k	0.301	896	ok
deepseek-v4-pro	low	100	1/1	92k	2k	0.128	100	ok
deepseek-v4-pro	medium	100	1/1	253k	8k	0.316	351	ok
deepseek-v4-pro	high	39	3/6	999k	36k	0.188	2074	ok
gpt-5.3-codex	low	100	2/2	177k	7k	0.470	149	ok

MODEL	BAND	SCORE	TASKS	IN TOK	OUT TOK	COST $	LAT (S)	PIN
gpt-5.3-codex	medium	100	4/4	402k	15k	1.081	428	ok
qwen3-coder-next	low	75	1/1	92k	2k	0.153	58	ok
qwen3-coder-next	medium	43	0/1	162k	4k	0.019	162	ok
qwen3-coder-next	high	22 *	0/1	4k	1k	0.000	75	FALLBACK

* = pin fell back (run invalid; row excluded from findings). Tokens are cumulative across the worker's full-history loop. Cost is raw (pre-cache) LLM spend — see caching note below.

Prompt caching (per provider · model, from per-call telemetry)

PROVIDER · MODEL	INPUT TOK	CACHE-READ TOK	~HIT
deepseek · deepseek-v4-flash-native	3.99M	3.70M	93%
openrouter · codestral-2508	407k	372k	91%
openrouter · devstral-2	1.39M	1.26M	91%
openrouter · gpt-5.3-codex	2.44M	2.21M	91%
openrouter · grok-4.20	12.88M	11.68M	91%
anthropic · claude-sonnet-4-6	294k	2.47M	89%
openrouter · minimax-m2.7	1.80M	1.57M	88%
openrouter · qwen3-coder-next	203k	174k	86%
openai · gpt-5.4	2.28M	1.88M	82%
openrouter · kimi-k2.7-code	1.22M	869k	71%
openrouter · deepseek-v4-pro	1.20M	837k	70%
openrouter · glm-5.2	1.08M	604k	56%
gemini · gemini-3.5-flash	4.85M	2.47M	51%
openrouter · minimax-m3	1.25M	143k	11%
openrouter · claude-sonnet-4-6 (fallback artifact)	3.48M	0	0%

> **Telemetry gap:** the /api/runs/{id}/cost endpoint stores only raw input/output and **drops cache-read tokens**, so the "Cost $" column above is a worst-case (pre-cache) figure — true post-cache spend is materially lower for the ~50–91% cached models. (Known W0 follow-up.) **minimax-m3 caches weakly (~11%)**, compounding its other issues.

Recommendations

- **Promote deepseek-v4-flash as the default cheap-tier coding model.** Best price/performance by a wide margin; HIGH-capable; native key (no gateway tax). Strong candidate for FREE/utility routing.
- **Per-model prompt tuning — targets locked: devstral-2** and **qwen3-coder-next**, reinforcing the test-writing directive. Data-backed (both are prompt-fixable, not capability-limited). Re-measure after.
- **Do not route HIGH-complexity work to** deepseek-v4-pro / glm-5.2 / minimax-m2.7 — they collapse on multi-task specs despite strong low/medium scores.
- **Fix the cache-read telemetry gap** so cost dashboards/estimates reflect true post-cache spend (presently overstated).

Methodology & integrity

- **Real production pipeline.** Each cell = one full V2 BUILD run (classify → plan → worker → reviewer → score) in PROTOTYPE mode, model pinned per-run via RunModelOverrides (ENTERPRISE).
- **Pin verified, not assumed.** Every gateway row was checked against the actual per-call RunCostTracker log (pinOK); a smoke pass first caught two real bugs — a require_parameters regression that 404'd valid OpenRouter models into a silent claude-sonnet fallback, and three non-existent Fireworks slugs — both fixed before the eval.
- **Dropped before eval:** devstral-small (0 tool endpoints anywhere), qwen3-coder-480b (= existing qwen3-coder), 3 invalid Fireworks slugs.

- **Incomplete (2 cells):** gpt-5.3-codex/high and qwen3-coder-next/high — OpenRouter account hit its $50 credit limit (42× 402) mid-run. Both are OpenRouter-only (no alternate route); they need a credit top-up to finish.
